Behavioral Analysis Framework

Objective

Convert observable behavioral signals into recruiter-friendly insights while maintaining privacy and fairness.

Step 1 — Capture Signals

Input signals include:

- Eye focus
- Head stability
- Facial engagement
- Attention patterns

Step 2 — Normalize Signals

Convert all raw values into a normalized range between **0 and 1**.

Example

Raw Value	Normalized
High	1.0
Medium	0.5
Low	0.2

Step 3 — Pattern Detection

Behavioral patterns detected:

- Focus
- Distraction
- Nervous gestures
- Engagement

Step 4 — Weighted Scoring

Signal	Weight
Eye Focus	30%
Head Stability	20%
Engagement	30%
Low Distraction	20%

Formula

```
Behavior Score =
Eye Focus × 0.30 +
Head Stability × 0.20 +
Engagement × 0.30 +
(1 – Distraction) × 0.20
```

Step 5 — Generate Insights

The system produces:

- Behavior Score
- Focus Level
- Engagement Level
- Risk Level

Example

```
{
    "behavior_score": 78.5,
    "focus_level": "High",
    "engagement": "Strong",
    "risk": "Low Risk"
}
```

Behavioral Levels

Score	Level
85–100	Highly Focused
70–84	Good Engagement
50–69	Moderate

Score	Level
Below 50	Distracted

Ethical Guidelines

- No facial recognition
- No identity storage
- Metadata only
- Candidate consent required
- Transparent scoring
- Explainable outputs

Benefits

- Improves interview evaluation
- Supports recruiter decisions
- Privacy preserving
- Easy integration

Future Enhancements

- Gesture recognition
- Blink detection
- Posture analysis
- Real-time attention monitoring